

MPOS User Guide: Estimator

Power Application Controller®



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Table of Contents

1 Description	3
2 Firmware Implementation.....	4
2.1 Estimator Module.....	4
2.2 Startup Sequence.....	5
2.3 Functions	6
2.4 Inputs.....	6
2.5 Outputs	6
3 Firmware Configuration.....	7
3.1 Selecting Estimator Mode	7
3.2 Compile Options and Parameters	7
3.3 GUI Parameters.....	8
4 Tuning Procedure.....	11
4.1 Entering GUI Parameters	11
4.2 Tuning Align&Go Startup.....	11
4.3 Tuning Angle Estimation.....	13
4.3.1 Background & Considerations	13
4.3.2 Phase Current Method.....	13
4.3.3 Efficiency Measurement Method	15
Contact Information.....	16
Important Notice	16

1 DESCRIPTION

The Estimator software module is one of the selectable MPOS (Motor Position) options available to the user for motor control using the Qorvo FOC firmware. When selected, the Estimator module uses the motor phase currents and voltages to calculate the estimated rotor position.

This sensor-less method of estimating the rotor position is typically used in applications where:

- 1) The motor does not need to operate at very low speeds which may be below the capability of the Estimator module.
- 2) Cost reduction is desired by eliminating the need for expensive position sensors such as hall sensors or QEP encoders

The FOC firmware uses a Luenberger based observer to estimate the motor position and speed in sensorless PMSM applications.

2 FIRMWARE IMPLEMENTATION

2.1 Estimator Module

The Estimator module uses motor phase currents and voltages to estimate the rotor position and speed as outlined in the following general steps:

- 1) Scaled phase currents and voltages are fed to the observer which estimates the angle using a motor model.
- 2) The estimated speed is then calculated from the calculated motor angle and a low-pass filter is applied. If “ENABLE_PLL_ANGLE_CALCULATION” is enabled, the motor angle and speed will be retrieved from PLL module.
- 3) Angle correction is applied to the calculated angle with two components: a constant offset component and a speed-dependent component.

Below is a block diagram that describes the estimator.

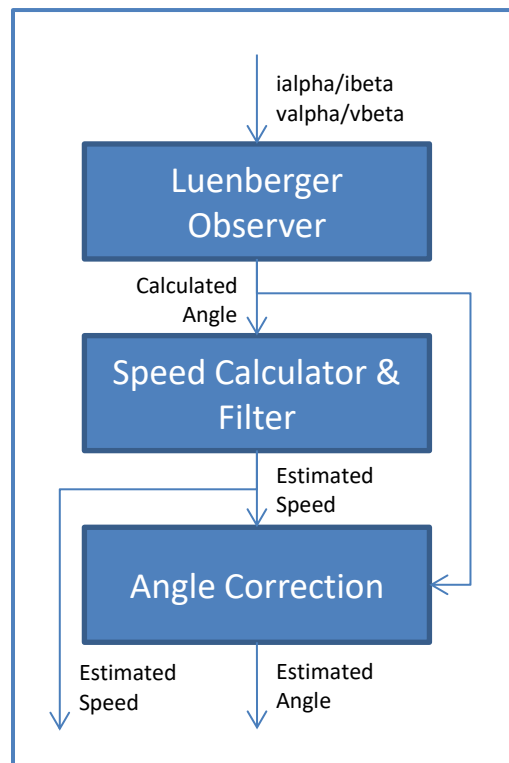


Figure 1: Estimator Module Block Diagram

2.2 Startup Sequence

Because the estimator uses the motor BEMF information, it can only estimate the motor position and speed when the motor is already in motion above a certain speed. Therefore starting the motor from zero-speed needs to be handled differently. The startup method can start the motor by driving the motor into motion without having any actual speed and rotor position information. During this time the speed of the drive vector is ramped to reach a speed where the estimator can run properly. When this speed is reached, the firmware switches over to using the estimated - instead of the forced - position and speed information. At this point the motor is now running with closed loop control.

The sequence followed during the motor startup process from zero-speed to closed loop operation is demonstrated in Figure 2.

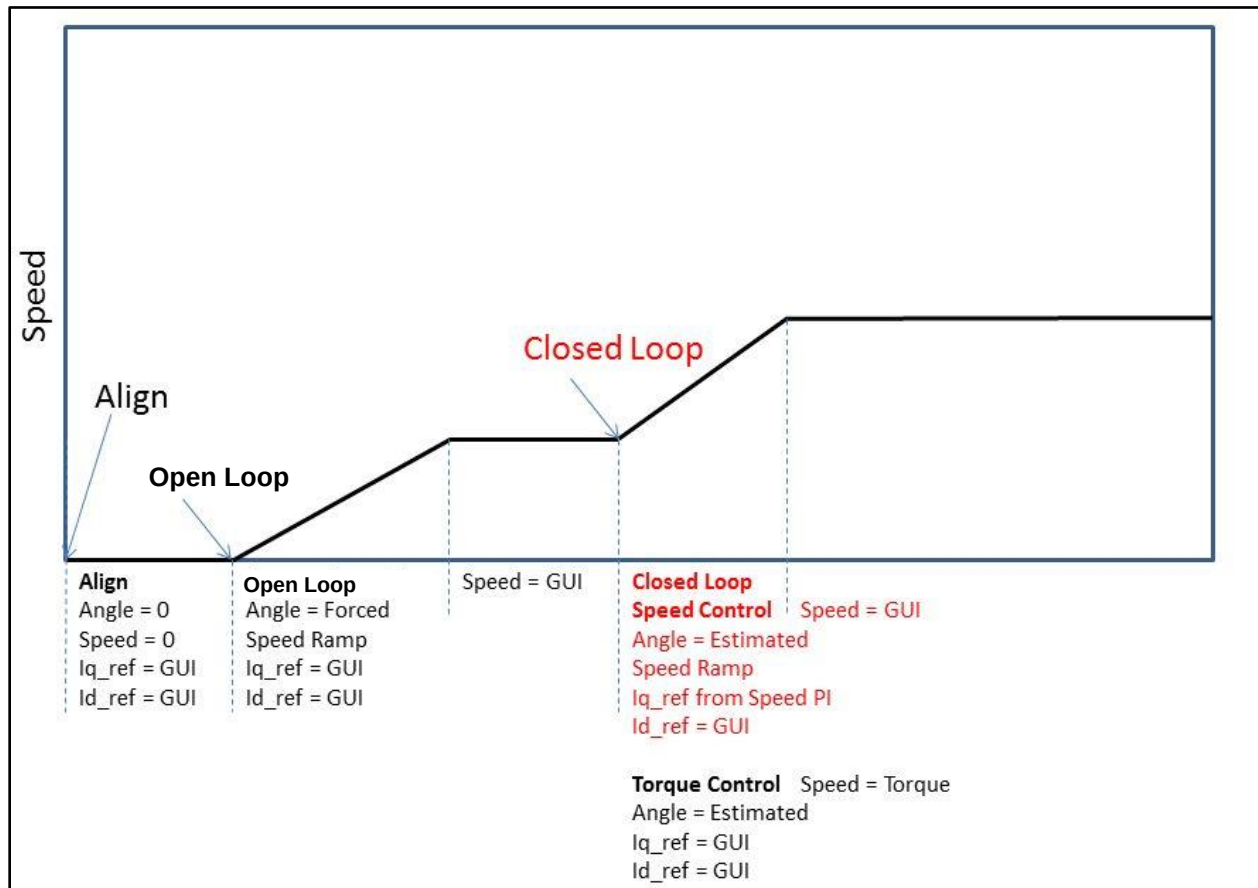


Figure 2: Startup Sequence for Estimator with Align&Go

2.3 Functions

The FOC firmware calls the following functions related to the estimator:

Function Name	Description
lbg1_prepare_inputs	Reads ialpha/ibeta and valpha/vbeta values from FOC and passes them to the estimator. Valpha/vbeta are passed through a shift register to account for firmware control loop delays.
lbg1_process_voltage_scale	Constantly updates the normalized motor model with the current bus voltage scale factor to account for dynamic bus voltage.
lbg1_update_state	Initializes the estimated angle and speed when doing a transition from SIM to estimator.
lbg1_check_transition_ready	Used to check if the estimator has completed its transition period when transitioning from SIM or from Align&Go startup to estimator.
lbg1_calculate_coefficients	Used to calculate internal estimator constants when GUI parameters are entered or changed.
lbg1_switch2cl	Starts the estimator transition period when doing a transition from SIM or from Align&Go startup to estimator.
lbg1_statemon	Executes the transition from SIM or from Align&Go startup to estimator. Runs at the application layer (systick) frequency.
lbg1_init	Resets the estimator internal variables when starting the motor.
lbg1_estimator	Estimates the motor position and speed based on motor current and voltage.

2.4 Inputs

The Luenberger estimator receives the following inputs from the FOC algorithm:

Input Variable Name	Format	Units
ialpha1	Q14 (1.0 = 16384)	Normalized current [-1, 1]
ibeta1	Q14 (1.0 = 16384)	Normalized current [-1, 1]
valpha1	Q14 (1.0 = 16384)	Normalized voltage [-1, 1]
vbeta1	Q14 (1.0 = 16384)	Normalized voltage [-1, 1]

The following normalization factors define the relationship between the normalized and de-scaled current and voltage values:

- $\text{descale_current_factor} = 1.25 \text{ (V)} / \text{shunt_resistor_value (R)} / \text{diff_amp_gain}$
- $\text{descale_voltage_factor} = \text{DC_bus_voltage (V)} * \sqrt{3}$

2.5 Outputs

The FOC algorithm receives the following outputs from the Luenberger estimator:

Output Variable Name	Format	Units
estimated_angle	Q16 (1.0 = 65536)	rad $[-\pi, \pi]$
est_spd radps	Q16 (1.0 = 65536)	rad/sec
est_spd_scaled_q14	Q14 (1.0 = 16384)	Normalized speed [-1, 1]
est_spd_scaled_q28	Q28 (1.0 = 268435456)	Normalized speed [-1, 1]

3 FIRMWARE CONFIGURATION

3.1 Selecting Estimator Mode

The Estimator module is enabled in “config_app.h” by selecting one of the following MPOS mode definitions depending on the behavior desired:

- 1) **#define** MPOS_SELECT MPOS_LBG1_START_ALIGNGO
This MPOS option starts the motor using Align&Go and switches to sensorless mode when the speed is above a user-configurable threshold.

- 2) **#define** MPOS_SELECT MPOS_LBG1_START_HALL
This MPOS option runs in interpolated hall sensor mode to start the motor and switches to sensorless mode when the speed is above a user-configurable threshold.

(Please refer to the Hall Sensor MPOS guide for more information on this mode selection.)

- 3) **#define** MPOS_SELECT MPOS_LBG1_START_QEP
This MPOS option runs in QEP sensor mode to start the motor and switches to sensorless mode when the speed is above a user-configurable threshold.

3.2 Compile Options and Parameters

The following compile options are available in “config_app.h” to configure the Estimator module:

ENABLE_OPEN_LOOP_NONLINEAR_RAMP

Apply two ramping rates on the open loop frequency ramp to speed up the motor.

ENABLE_VOLTAGE_BASED_OPEN_LOOP_STARTUP

Apply the fixed voltage during open loop ramping. If it is disabled, the current controlled open loop ramping will be applied.

ENABLE_OPEN_LOOP_RAMP_NONZERO_START

Start a non-zero frequency to ramp open loop frequency.

ENABLE_ANGLE_CORRECTION

Apply angle correction to calculated angle consisting of a constant offset component and a speed-proportional component.

ENABLE_PHASE_COMPENSATION

Apply automatic estimated angle correction based on the phase delay of the observer due to observer bandwidth.

ENABLE_PLL_ANGLE_CALCULATION

Apply PLL based angle and speed estimation for the calculated back EMF. **SWITCHOVER_SPD_HALLEST_HYST_PCT_Q16**

Hysteresis band for switching from Estimator to Hall mode as a percentage of user-defined switchover speed.

ENABLE_ANGLEDIFF_SPEED_CHECK

Check that the angle and speed are within an error threshold before switching between estimator and hall modes. If the check fails, warning is reported. When enabled, the following options can be configured:

ANGLEDIFF_SPEED_ERROR_PCT_Q16

Allowed speed error band as a percentage of the switchover speed and entered in decimal format. Recommended value = 0.1.

ANGLEDIFF_ANGLE_ERROR_DEG_Q12

Allowed angle error band in radians. Recommended value = $\pi/10$.

ENABLE_OLSPEED_ANGLEDIFF_TIMEOUT

Enables timeout for anglediff and speed check. If the check fails for the specified timeout period, the motor is disabled. When enabled, the following option can be configured:

ANGLEDIFF_TIMEOUT_SEC

Timeout period defined in seconds. Recommended value = 2.

ENABLE_ANGLEDIFF_AUTORANGE

Enables open-loop AngleDiff auto ranging when switching from Align&Go to Estimator mode. Enabling this function dramatically improves startup reliability. It works by automatically ramping `id_ref` in order to converge the forced angle and the estimated angle for a smooth switchover. When enabled, the following option can be configured:

ANGLEDIFF_AUTORANGE_INC_METHOD1_PCT

Defines the rate at which auto-ranging increments `id_ref` when converging the forced angle and the estimated angle. The rate is expressed as a percentage of Startup current and is incremented every top-level control period (normally 10ms). Recommended value = 0.05.

3.3 GUI Parameters

The following GUI parameters are required by the Estimator module to construct the motor model and estimate the rotor position:

Parameter Name	Units	Description
BW	Hz	Observer Bandwidth for second order low pass filter system
DF	-	Observer Damping Factor for second order low pass filter



		system
Fc-speed	Hz	Speed filter cutoff frequency
Angle Correction Offset	Degrees	Constant angle offset applied to the angle calculated by the observer
Angle Correction Factor	Degrees / Hz	Gain factor used to apply angle correction that is proportional to the motor speed
L, R	uH, Ohms	Motor winding inductance and resistance. Used to calculate motor model constants for observer (fs, gs).
Control Freq	Hz	Used to calculate speed from the angle obtained from the observer
Nominal Speed	Hz	Used to automatically enter the observer bandwidth when the AutoBW box is checked.
Max Speed	eHz	Used to calculate the normalized estimated speed

LBG1 Estimator

BW (Hz)

1200.0

Kpe

DF

0.7071

Kpi

AutoBW

☐

Fc-speed (Hz)

70

Angle Correction

Offset (deg)

0.00

Factor (deg/Hz)

0.0000

R

W

Figure 3: Estimator Tuning Section

Config

Tuning

Features

Plots

Motor Config

Ld (uH)

25

Lq (uH)

35

Ralpha (Ohm)

0.0413

pole pairs

4

Kt (Nm/Amp)

0.0140

J (Nm/a)

0.00002

Nominal Speed (Hz)

600.00

App Config

Speed (Hz)

1000.00

Power (W)

400.00

Current (A)

20.00

Negative Torque (A)

0.00

OC Limit (A)

25.00

CL coast timeout (s)

1.00

Control Config

PWM Freq (Hz)

24000

Control Divider

1

Control Freq 1/dt (Hz)

24000

Hardware Config

OC Rsense (mR)

Iuvw Rsense (mR)

Diff Amp gain (x)

4

Current Scale (A)

Vdc Ratio

Vphase Ratio

Vdc Max (V)

Figure 4: Config Panel

4 TUNING PROCEDURE

In order to tune the Estimator module, please follow the following steps:

4.1 Entering GUI Parameters

- 1) Enter a value for the following Estimator GUI parameters:
 - a) BW : This is the observer bandwidth. By default this value should be set to anywhere between 1-4x Nominal_Speed. Also it should be set lower than $1/15 \times \text{control_frequency}$. The initial recommendation is to start with 2x Nominal_Speed unless it is too high for the control_frequency. If the AutoBW box is checked, the bandwidth is automatically filled in with 2x Nominal_Speed.
 - b) DF : This is the damping factor for the observer. The default value is $1/\sqrt{2}$ which guarantees stability in a second order system.
 - c) Fc-speed (Hz): This is the cutoff frequency for the speed filter. Setting this cutoff frequency too low can cause instability in the speed response. Setting it too high can result in excessive current waveform noise as well as audible noise. This frequency needs to be selected based on the acceleration rates for each particular application. Some guideline examples are listed below:
 - Drone motors: 70Hz due to very fast acceleration rates
 - Compressors: 20Hz
 - Ceiling fans: 5Hz due to very low acceleration rates
 - d) Angle Correction Offset and Gain Factor: These should initially be set to 0. They will be adjusted later to optimize the estimated angle.
- 2) Enter a value for the following Align&Go GUI parameters:
 - a) Align time : 3 seconds
 - b) "Id ref OL" : 0
 - c) "I total ref OL" : Set starting current to about 20% of max continuous motor current (A)
 - d) OL speed ramp : 5% of maximum electrical speed (Hz/s)
 - e) OL speed setpoint : About 10% of maximum electrical speed (Hz)
 - f) Check that the Close Loop checkbox is not selected.

4.2 Tuning Align&Go Startup

- 3) Determine the Align&Go starting current value that can spin the motor reliably through the open-loop speed ramp to the switchover speed:
 - a) Enable the motor. The motor should go through the align state before it starts turning. Observe the behavior of the motor and check against the possible scenarios below:
 - If the motor is turning but it is doing so roughly and you need to apply a lot of force with your hand to stop it, reduce the "I total ref OL" .
 - If the motor is not turning properly, try to help the motor to turn with your hand. If it starts turning smoothly, then increase the "I total ref OL" value.
 - If the motor turns but it stops turning when you apply very slight resistance to the rotor, then increase the "I total ref OL" value.
 - If the motor starts turning initially but then stalls as the speed command is increasing, then increase the "I total ref OL" value.
 - Note that increasing the open-loop acceleration rate will require a higher starting current to prevent motor stalling while speed is ramping.
 - b) After these steps, the motor should be able to reach the open-loop speed setpoint smoothly every time that the motor is enabled. If the motor stalls occasionally, the guidelines above need to be revisited.

- 4) Once the Align&Go is tuned properly, the Close Loop checkbox can be selected so that the firmware proceeds to switch to Estimator mode. If the speed controller is not tuned yet, it is possible to proceed in Torque Mode instead of Speed Mode. The switch to Estimator mode is shown in Figure 5 below.

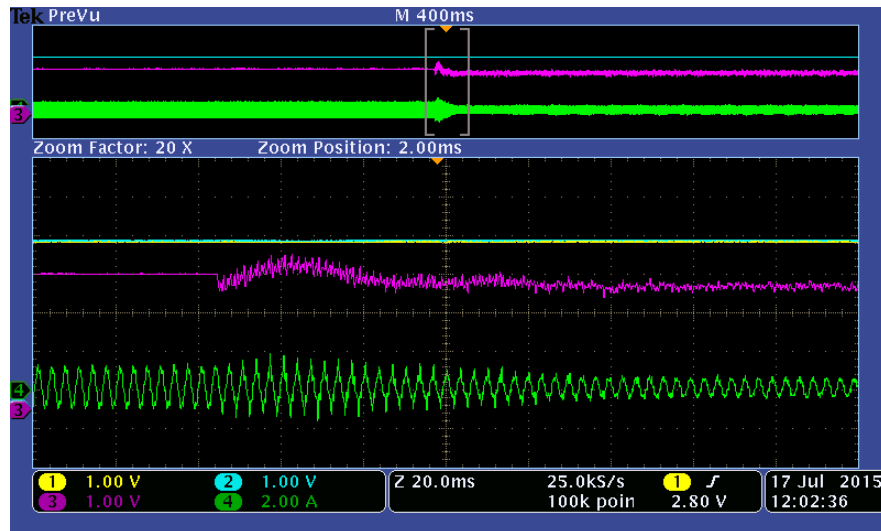


Figure 5: Successful switch from Align&Go to Estimator mode

4.3 Tuning Angle Estimation

After a successful transition from Align&Go to Estimator mode, estimated angle must be tuned in order to ensure that the angle generated by the observer represents the actual rotor angle as closely as possible. This process will maximize the efficiency and performance of the system.

4.3.1 Background & Considerations

- 1) In order to maximize efficiency, what really matters is that the estimated angle is correct and maintains the most optimal value to drive the motor such that the motor current gets converted entirely into torque and not flux.
- 2) There are factors that could introduce error into the angle estimation for which it may be necessary to compensate manually with angle correction:
 - a) Error between the L/R values entered in the GUI and the actual values.
 - b) Difference between L_d and L_q (Only L_q is considered in the observer motor model and L_d is assumed to be same as L_q .)
 - c) Parasitics such as wire inductance/resistance
 - d) FET characteristics such as $R_{ds(on)}$
- 3) Estimator angle optimization can be done in a couple of ways:
 - a) Phase current method
 - b) Efficiency measurement method

4.3.2 Phase Current Method

This step assumes that the motor being tuned is not perfectly sinusoidal. As a result the non-sinusoidal characteristics of the motor can be observed in the phase current. By looking at the current waveform, we can then determine if the angle is advanced or delayed relative to the optimal position.

Please follow the following steps:

- 1) Set motor speed to the switchover speed for Align&Go.
- 2) Observe the phase current waveform and compare with the scope shots shown below.
- 3) Depending on the phase current waveform adjust "Angle Correction Offset" according to the following guidelines to get to the optimal angle:
 - If angle is advanced, delay the angle by decreasing Offset
 - If angle is delayed, advance by increasing Offset
- 4) Increase the speed to 50% of top speed and adjust "Angle Correction Gain" using the same guidelines from step #3 above.
- 5) Increase the speed to 75% and repeat step #4.

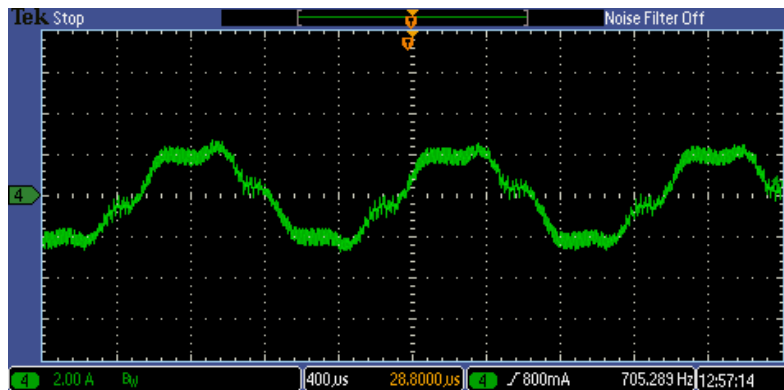


Figure 6: Phase current of delayed angle

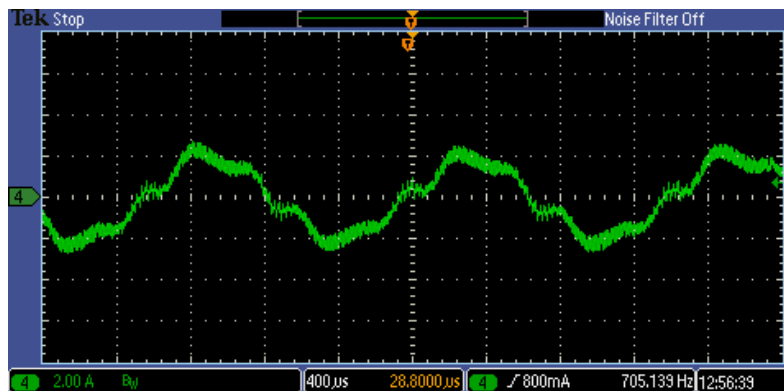


Figure 7: Phase current of advanced angle

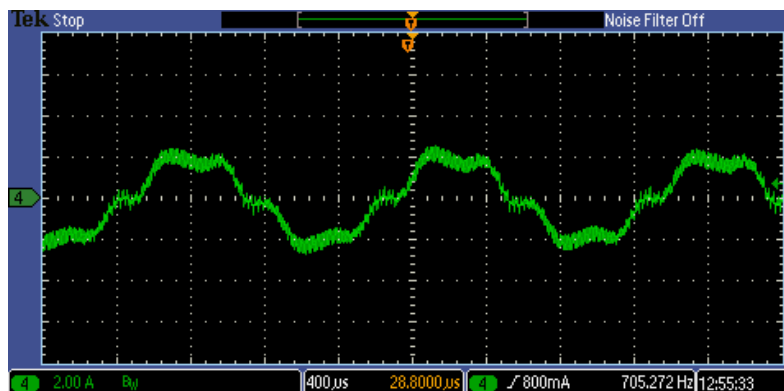


Figure 8: Phase current of optimal angle

Note that at higher speeds the non-sinusoidal characteristics become less noticeable as shown in Figure 9.

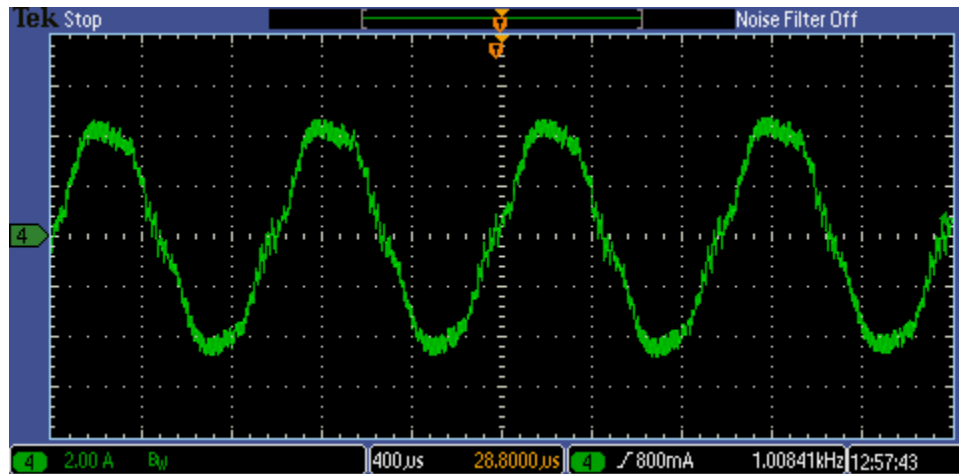


Figure 9: Phase current of optimal angle at high speed

4.3.3 Efficiency Measurement Method

Confirm that angle estimation is at the optimal point by using the efficiency/power method:

- 1) Set motor speed to the switchover speed for Align&Go.
- 2) Measure 3-phase motor power and the supply power to calculate efficiency. If only supply efficiency is available then just the supply power can be used.
- 3) Adjust “Angle Correction Offset” to minimize the supply power or maximize the efficiency.
- 4) Increase the speed to 50% of top speed and adjust “Angle Correction Gain” to minimize the supply power or maximize the efficiency.
- 5) Increase the speed to 75% and repeat step #4.

Note: If the non-sinusoidal characteristics of the motor current are not very obvious, then the efficiency measurement method would be the only way to calibrate the estimated angle.



CONTACT INFORMATION

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